

Synthetic EV Battery Health

Week-9.1: Machine Learning and Data Science Capstone Project

- **About Dataset**

Synthetic EV Battery Health Dataset

A **Synthetic EV Battery Health Dataset** is an artificially generated dataset that simulates the operating behaviour and degradation of electric vehicle (EV) batteries. It is designed to mimic real-world battery data while avoiding privacy, proprietary, or data collection constraints. Such datasets are commonly used for machine learning, battery health prediction, predictive maintenance, and academic research.

Columns	Description
Battery_ID	Unique identifier for each battery
Cycles	Number of charge/discharge cycles
Voltage	Battery terminal voltage (V)
Current	Charging/discharging current (A)
Temperature	Operating temperature (°C)
Capacity	Remaining battery capacity (Ah)
Internal Resistance	Battery resistance (mΩ)
State of Charge (SOC)	Current charge level (%)
Depth of Discharge (DoD)	Percentage of battery discharged
C-rate Charge/discharge rate	Charge/discharge rate
Ambient Temperature	External temperature (°C)
Battery Age	Age of the battery (days or months)
SOH	State of Health (%) – target variable
RUL	Remaining Useful Life (cycles or days)

Purpose:

The primary objectives of a synthetic EV battery health dataset are to:

- Predict State of Health (SOH).
- Estimate Remaining Useful Life (RUL).
- Detect battery degradation patterns.
- Support predictive maintenance.
- Train and evaluate machine learning models.